Sampling Methods & Bias

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Identifying sampling methods**

A school assigns each student a number from 1 to 800, then uses a random number generator to select 40 numbers, surveying those students. Identify this sampling method.

This is a simple random sample (SRS): every student has an equal chance of being selected, and every group of 40 students has an equal chance of being the chosen sample.

2 **[Foundational] • Target: Identifying sampling methods**

A researcher divides a school into grade levels (9th, 10th, 11th, 12th), then takes a simple random sample of 20 students from each grade. Identify this sampling method and explain why it might be preferred over an SRS of the whole school.

This is stratified random sampling. It might be preferred because it guarantees representation from every grade level, which is useful if opinions or characteristics vary meaningfully by grade — an SRS of the whole school could, by chance, under-represent or over-represent a particular grade.

3 **[Foundational] • Target: Cluster sampling and systematic sampling**

A researcher randomly selects 5 entire homeroom classes out of 40 in a school, and surveys every student in those 5 classes. Identify this sampling method, and contrast it with stratified sampling.

This is cluster sampling: entire naturally-occurring groups (homerooms) are randomly selected, and every member of the selected groups is surveyed. This differs from stratified sampling, where the researcher samples from every group, rather than sampling only a few whole groups and surveying everyone within them.

4 **[Foundational] • Target: Cluster sampling and systematic sampling**

A quality-control inspector checks every 25th item coming off a production line. Identify this sampling method.

This is systematic sampling: individuals are selected at a fixed interval (every 25th item) from an ordered list or sequence, typically starting from a randomly chosen point.

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5 [Foundational] • Target: Sources of bias

A survey about a new school policy is voluntarily filled out online, and only 12% of students respond. Identify the type of bias most likely present, and explain why the results may not represent the whole student body.

This is voluntary response bias: students with strong opinions (especially strongly negative or strongly positive ones) are more likely to take the time to respond, so the 12% who responded are not representative of the full student body's range of opinions — moderate or indifferent students are underrepresented.

6 [Foundational] • Target: Sources of bias

A phone survey about internet usage habits is conducted using only landline phone numbers. Identify the type of bias this introduces.

This introduces undercoverage bias: households that only use cell phones (increasingly common, especially among younger people) are systematically excluded from the sampling frame, so the sample cannot represent the full population of interest.

7 [Foundational] • Target: Wording and response bias

A survey question reads: "Don't you agree that the wasteful new cafeteria policy should be reversed?" Explain why this question is problematic, and rewrite it in a neutral form.

The question is leading — the word "wasteful" and the phrasing "don't you agree" push respondents toward a particular answer, biasing the results. A neutral rewrite: "Do you support or oppose the new cafeteria policy?"

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Wording and response bias

Explain why respondents might not answer a sensitive question (such as about illegal activity) truthfully in a face-to-face interview, and name this type of bias.

This is response bias: when asked sensitive questions directly by an interviewer, respondents may feel social pressure to give a more socially acceptable answer rather than the truthful one, especially without the anonymity that a written or online survey might provide.

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9 **[Intermediate]** • **Target: Comparing sample size and bias**

Explain why increasing the sample size does not fix a biased sampling method, using the voluntary response survey from earlier as an example.

Bias is a systematic flaw in how the sample was selected, not a matter of the sample being too small — collecting responses from a larger number of self-selected volunteers would still overrepresent people with strong opinions. Only fixing the sampling method itself (e.g. switching to a simple random sample of all students) removes the bias; a larger biased sample is still biased, just with a more precise estimate of the wrong quantity.

10 **[Intermediate]** • **Target: Designing a sampling plan**

A city has 4 distinct neighborhoods with very different income levels, and the mayor wants to survey residents about a new tax proposal in a way that fairly represents each neighborhood. Recommend a sampling method, and describe how it would be carried out.

Stratified random sampling is recommended: divide the population into the 4 neighborhoods (strata), then take a simple random sample of residents from each neighborhood, proportional to (or intentionally balanced across) each neighborhood's population. This ensures every income group is represented, rather than relying on chance to include them proportionally as an SRS of the whole city might not.

11 **[Intermediate]** • **Target: Convenience sampling and its pitfalls**

A researcher stands outside a gym and surveys the first 50 people who walk out about their exercise habits, intending to represent the whole city's exercise habits. Identify this sampling method and explain the flaw in generalizing its results to the whole city.

This is convenience sampling: the researcher selects whoever is easiest to reach rather than using any random selection process. The results cannot be generalized to the whole city because people at a gym are not representative of the general population — they are systematically more likely to exercise regularly than the average city resident, biasing the sample toward higher reported exercise habits.

12 **[Intermediate]** • **Target: Comparing sampling methods side by side**

A national retailer wants to estimate average customer satisfaction across all 500 of its stores. Compare how a cluster sample (randomly selecting 20 whole stores and surveying all customers there) and a stratified sample (grouping stores by region, then sampling customers from every region) would each be carried out, and state one advantage of each approach.

Cluster sampling: randomly select 20 stores, then survey all customers at those 20 stores — cheaper and more logistically convenient since fieldwork is concentrated in fewer physical locations. Stratified sampling: divide the 500 stores into regions (strata), then sample customers from every region — guarantees geographic representation across all regions rather than relying on the 20 randomly chosen stores to happen to cover every region well.

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13 [Intermediate] • Target: Identifying the population and sample

A university wants to know the average commute time for all 30,000 enrolled students, and surveys a random sample of 400 students. Identify the population and the sample in this scenario.

Population: all 30,000 enrolled students at the university. Sample: the 400 students who were randomly selected and surveyed.

14 [Intermediate] • Target: Identifying the population and sample

A national news outlet runs a text-in poll during a live broadcast, asking viewers to text "YES" or "NO" on a controversial policy question, and reports the results as representing "public opinion." Identify every source of bias you can find in this scenario, and explain why the reported percentages should not be trusted as an accurate reflection of the general public's views.

This scenario has at least two compounding sources of bias: (1) undercoverage — only viewers of that specific broadcast, who own a phone capable of texting and chose to watch that program, are even eligible to respond, excluding large portions of the public; (2) voluntary response bias — among those eligible, only viewers with strong opinions (often skewed toward one side) are motivated to actually text in. Together, these make the reported percentages reflect the opinions of a self-selected, unrepresentative subgroup, not the general public.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A hospital wants to estimate the average patient satisfaction score across its 12 different departments (e.g. cardiology, pediatrics, emergency), which are expected to have very different typical satisfaction levels. Describe, step by step, how you would design a stratified random sample to estimate overall patient satisfaction, and explain why a simple random sample of all patients might be less informative for comparing satisfaction across departments.

Step 1: treat each of the 12 departments as a separate stratum. Step 2: take an independent simple random sample of patients from within each department. Step 3: combine the department-level samples into the overall estimate (optionally weighting by each department's size). This guarantees every department is represented and allows department-by-department comparisons. A simple random sample of all patients, by contrast, might by chance draw very few patients from a smaller department, making it hard to estimate or compare that department's satisfaction reliably.

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16 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A researcher wants to estimate the average time it takes customers to be served at a chain of 200 coffee shops nationwide. Due to budget constraints, the researcher can only visit a small number of physical locations. Describe how a cluster sample could be carried out in this situation, and explain a key risk of cluster sampling that would not apply if the researcher instead used a stratified sample based on region.

A cluster sample would involve randomly selecting a small number of entire coffee shops (say, 15 out of 200) and measuring service time for every customer at those 15 locations. A key risk is that if certain regions or store types happen to differ meaningfully in typical service time, the randomly selected 15 shops could, purely by chance, fail to represent that variation well (e.g. all 15 could happen to be busy urban locations). A stratified sample based on region would guarantee representation from every region by design, rather than leaving it to chance which regions end up represented.

17 [Challenging] • Target: Wording and response bias • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A school district with 12 high schools wants to sample 300 teachers. Compare: (a) Stratified sampling (taking an SRS of 25 teachers from each of the 12 schools) vs. (b) Cluster sampling (randomly selecting 2 schools and surveying all 150 teachers in each). Discuss sampling variability and logistical practicality.

(a) Stratified Sampling: Guarantees representation from every high school. Because schools may have differing cultures and student demographics (homogeneous within schools, heterogeneous between schools), stratifying reduces sampling variability between strata, providing a more precise district-wide estimate. (b) Cluster Sampling: Highly practical and cost-effective because surveyors only visit 2 physical campuses. However, if individual schools differ substantially from one another, cluster sampling introduces much higher sampling variability and may yield a non-representative sample if the two chosen schools happen to be atypical.

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18 [Challenging] • Target: Comparing sample size and bias • Cross-Lesson: 1.01 Measures of Center & Spread

A town council is conducting a survey about a proposed new park. They decide to use a stratified random sample to ensure representation from different neighborhoods. The total population of the town is 45,000. The town is divided into three neighborhoods: Northside (18,000 residents), Eastside (15,000 residents), and Westside (12,000 residents). The council plans to survey a total of 900 residents. If the sample is to be proportional to the population of each neighborhood, how many residents should be surveyed from the Northside neighborhood?

The problem requires us to determine the number of residents to be sampled from the Northside neighborhood based on proportional allocation. 1. Calculate the proportion of the total population that resides in the Northside neighborhood: Proportion (Northside) = (Population of Northside) / (Total Population) Proportion (Northside) = 18,000 / 45,000 2. Simplify the proportion: Proportion (Northside) = 18 / 45 = 2 / 5 3. Calculate the number of residents to be sampled from the Northside by multiplying this proportion by the total sample size: Sample Size (Northside) = Proportion (Northside) * Total Sample Size Sample Size (Northside) = (2/5) * 900 4. Perform the multiplication: Sample Size (Northside) = 2 * (900 / 5) Sample Size (Northside) = 2 * 180 Sample Size (Northside) = 360 Therefore, 360 residents should be surveyed from the Northside neighborhood. Common errors could include incorrectly calculating the proportion or applying it to a sub-total rather than the total population and total sample size.

19 [Challenging] • Target: Designing a sampling plan • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A diagnostic medical screening test is 95% sensitive (positive if disease present) and 90% specific (negative if disease absent). In a population, 2% of people have the disease. Find the probability that a person who tests positive actually has the disease.

Let D = disease present, T^+ = positive test. $P(D) = 0.02$, $P(D^c) = 0.98$. Sensitivity: $P(T^+ \mid D) = 0.95$. Specificity: $P(T^- \mid D^c) = 0.90 \Rightarrow P(T^+ \mid D^c) = 0.10$. By the Law of Total Probability: $P(T^+) = P(D)P(T^+ \mid D) + P(D^c)P(T^+ \mid D^c) = (0.02)(0.95) + (0.98)(0.10) = 0.019 + 0.098 = 0.117$. By Bayes' Theorem: $P(D \mid T^+) = \frac{P(D \cap T^+)}{P(T^+)} = \frac{0.019}{0.117} \approx \mathbf{0.1624}$ (16.24%). Despite 95% sensitivity, only ~16.2% of positive testers have the disease due to low baseline prevalence.

20 [Challenging] • Target: Convenience sampling and its pitfalls • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$?

For a continuous uniform distribution over the interval $[a, b]$, the probability density function (PDF) is $f(x) = \frac{1}{b-a}$ for $a \le x \le b$, and 0 otherwise. In this case, $a = 2$ and $b = 8$, so the length of the interval is $b - a = 8 - 2 = 6$. The PDF is $f(x) = \frac{1}{6}$ for $2 \le x \le 8$. The probability $P(4 < X < 7)$ is the integral of the PDF from 4 to 7, or equivalently, the area under the PDF curve between 4 and 7. $P(4 < X < 7) = \int_4^7 \frac{1}{6} dx$ Since the PDF is constant, this is simply the height of the PDF multiplied by the width of the interval $[4, 7]$. Width = $7 - 4 = 3$. $P(4 < X < 7) = \frac{1}{6} \times 3 = \frac{3}{6} = \frac{1}{2} = 0.50$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying sampling methods

A factory assigns each worker a number from 1 to 1223, then uses a random number generator to select 70 numbers, surveying those workers. Identify this sampling method.

a. Convenience sample

b. **Simple random sample (SRS) ✓**

c. Stratified random sample

d. Cluster sample

Simple random sample (SRS). Every worker has an equal chance of being selected, and every possible group of 70 workers has an equal chance of being chosen — this is the defining feature of an SRS.

22 [Foundational] • Target: Cluster sampling and systematic sampling

A researcher randomly selects 3 entire delivery routes out of 33 at a shipping company, and surveys every member within those selected delivery routes. Identify this sampling method.

a. Stratified random sample

b. Simple random sample (SRS)

c. Systematic sample

d. **Cluster sample ✓**

Cluster sample. Whole naturally-occurring groups (delivery routes) are randomly selected, and every individual within a selected group is surveyed — this is cluster sampling.

23 [Foundational] • Target: Sources of bias

A researcher stands outside a farmers market and surveys the first 44 people encountered, intending to represent the general population. Identify this sampling method.

a. Systematic sample

b. Cluster sample

c. Simple random sample (SRS)

d. **Convenience sample ✓**

Convenience sample. The researcher selects whoever is easiest to reach rather than using any random selection process — this is convenience sampling.

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24 [Foundational] • Target: Wording and response bias

A researcher divides a hospital into groups (department A, department B, and department C), then takes a simple random sample of 72 patients from each group. Identify this sampling method.

a. Cluster sample

b. **Stratified random sample ✓**

c. Convenience sample

d. Simple random sample (SRS)

Stratified random sample. The population is divided into groups (strata), and an independent simple random sample is taken from every stratum — this is stratified random sampling.

25 [Foundational] • Target: Comparing sample size and bias

A researcher stands outside a train station and surveys the first 49 people encountered, intending to represent the general population. Identify this sampling method.

a. Cluster sample

b. Simple random sample (SRS)

c. **Convenience sample ✓**

d. Systematic sample

Convenience sample. The researcher selects whoever is easiest to reach rather than using any random selection process — this is convenience sampling.

26 [Foundational] • Target: Designing a sampling plan

A researcher divides a factory into groups (freshmen, sophomores, juniors, and seniors), then takes a simple random sample of 67 workers from each group. Identify this sampling method.

a. Cluster sample

b. Systematic sample

c. **Stratified random sample ✓**

d. Simple random sample (SRS)

Stratified random sample. The population is divided into groups (strata), and an independent simple random sample is taken from every stratum — this is stratified random sampling.

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27 [Foundational] • Target: Convenience sampling and its pitfalls

A quality-control inspector at a clinic checks every 32nd item coming off a production line, starting from a randomly chosen point among the first 32 items. Identify this sampling method.

a. Simple random sample (SRS)

b. Convenience sample

c. **Systematic sample ✓**

d. Cluster sample

Systematic sample. Individuals are selected at a fixed interval (every 32nd item) from an ordered sequence, after a random starting point — this is systematic sampling.

28 [Foundational] • Target: Comparing sampling methods side by side

A survey about a new dress code is voluntarily filled out online by residents, and only 20% respond. Identify the type of bias most likely present.

a. Wording bias (leading question)

b. Convenience sampling bias

c. **Voluntary response bias ✓**

d. Response bias (social desirability)

Voluntary response bias. Residents with strong opinions are far more likely to take the time to respond, so the 20% who responded are not representative of the full population's range of opinions.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Identifying the population and sample

A survey question reads: "Don't you agree that the wasteful dress code should be changed?" Identify the type of bias in this question.

a. Convenience sampling bias

b. Voluntary response bias

c. **Wording bias (leading question) ✓**

d. Response bias (social desirability)

Wording bias (leading question). The word "wasteful" and the phrasing "don't you agree" push respondents toward a particular answer, biasing the results.

30 [Intermediate] • Target: Probability Sampling Designs

A survey about a new policy is voluntarily filled out online by employees, and only 14% respond. Identify the type of bias most likely present.

a. Convenience sampling bias

b. Undercoverage bias

c. **Voluntary response bias ✓**

d. Response bias (social desirability)

Voluntary response bias. Employees with strong opinions are far more likely to take the time to respond, so the 14% who responded are not representative of the full population's range of opinions.

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31 [Intermediate] • Target: Selection Bias & Undercoverage

Respondents are asked directly, face-to-face, about tax evasion. Identify the type of bias likely to affect their answers.

a. Undercoverage bias

b. **Response bias (social desirability) ✓**

c. Wording bias (leading question)

d. Convenience sampling bias

Response bias (social desirability). When asked sensitive questions directly by an interviewer, respondents often give a more socially acceptable answer rather than the truthful one.

32 [Intermediate] • Target: Nonresponse and Response Bias

A survey about a proposed schedule change is voluntarily filled out online by employees, and only 8% respond. Identify the type of bias most likely present.

a. **Voluntary response bias ✓**

b. Response bias (social desirability)

c. Wording bias (leading question)

d. Convenience sampling bias

Voluntary response bias. Employees with strong opinions are far more likely to take the time to respond, so the 8% who responded are not representative of the full population's range of opinions.

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33 [Intermediate] • Target: Parameters vs. Statistics

Respondents are asked directly, face-to-face, about illegal drug use. Identify the type of bias likely to affect their answers.

a. Undercoverage bias

b. Convenience sampling bias

c. **Response bias (social desirability) ✓**

d. Voluntary response bias

Response bias (social desirability). When asked sensitive questions directly by an interviewer, respondents often give a more socially acceptable answer rather than the truthful one.

34 [Intermediate] • Target: Identifying sampling methods

A researcher surveys people at a train station and generalizes the results to the entire city's population. Identify the type of bias most likely present.

a. Voluntary response bias

b. Wording bias (leading question)

c. **Convenience sampling bias ✓**

d. Response bias (social desirability)

Convenience sampling bias. People found at a train station are not representative of the general population, since the location systematically attracts a particular type of person.

35 [Intermediate] • Target: Cluster sampling and systematic sampling

At a school with 35 homeroom classes, a researcher randomly selects 4 entire homeroom classes and surveys every individual within them. Identify this sampling method.

a. Systematic sample

b. Convenience sample

c. **Cluster sample ✓**

d. Simple random sample (SRS)

Cluster sample. Whole homeroom classes are randomly selected and every member within the selected homeroom classes is surveyed, without sampling from the homeroom classes that were not selected — this is cluster sampling, not stratified sampling.

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36 [Intermediate] • Target: Sources of bias

At a hospital with 60 wards, a researcher randomly selects 3 entire wards and surveys every individual within them. Identify this sampling method.

a. Simple random sample (SRS)

b. Convenience sample

c. **Cluster sample** ✓

d. Stratified random sample

Cluster sample. Whole wards are randomly selected and every member within the selected wards is surveyed, without sampling from the wards that were not selected — this is cluster sampling, not stratified sampling.

37 [Intermediate] • Target: Wording and response bias

A clinic has 25805 enrolled patients in total. A researcher surveys a random sample of 873 patients to estimate average commute time. Find the size of the population in this scenario.

a. 25705

b. 25805 ✓

c. 25905

d. 873

The population in this scenario is 25805 patients.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing sample size and bias • Cross-Lesson: 1.01 Measures of Center & Spread

A company has 44627 enrolled employees in total. A researcher surveys a random sample of 422 employees to estimate average commute time. Find the size of the population in this scenario.

a. 44727

b. 44627 ✓

c. 422

d. 44527

The population in this scenario is 44627 employees.

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39 [Challenging] • Target: Designing a sampling plan • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A pollster has a complete numbered list of all registered voters and wants every voter to have an exactly equal chance of selection. Recommend the most appropriate sampling method.

a. Systematic sample

b. Stratified random sample

c. **Simple random sample (SRS) ✓**

d. Convenience sample

Simple random sample (SRS) is most appropriate for this scenario.

40 [Challenging] • Target: Convenience sampling and its pitfalls • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A factory manager wants a cheap, logistically convenient way to check product quality by inspecting a few whole production batches out of many. Recommend the most appropriate sampling method.

a. Systematic sample

b. Stratified random sample

c. **Cluster sample ✓**

d. Simple random sample (SRS)

Cluster sample is most appropriate for this scenario.

41 [Challenging] • Target: Probability Sampling Designs • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Which sampling method gives every possible group of n individuals an equal chance of being chosen?

a. **Simple Random Sample (SRS) ✓**

b. Stratified Random Sample

c. Cluster Sample

d. Systematic Sample

By definition, a Simple Random Sample (SRS) of size n guarantees that every individual and every subset of n individuals has the exact same probability of selection.

42 [Challenging] • Target: Probability Sampling Designs • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A city divides its territory into 40 voting precincts. Five precincts are randomly chosen, and every resident in those 5 precincts is surveyed. What sampling method is this?

a. **Cluster sampling ✓**

b. Stratified sampling

c. Simple random sampling

d. Multistage convenience sampling

Dividing the population into heterogeneous clusters (precincts), randomly choosing whole clusters, and surveying all members within selected clusters is cluster sampling.

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43 [Challenging] • Target: Selection Bias & Undercoverage • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Can increasing the sample size from 100 to $10,000$ eliminate bias in a voluntary response survey?

a. No, a larger biased sample simply estimates the biased quantity more precisely ✓

b. Yes, large samples eliminate all forms of sampling bias by the Law of Large Numbers

c. Yes, provided $n \geq 1000$

d. Only if the population is normally distributed

Sample size reduces sampling variability (random error), but cannot fix systematic design flaws or bias.

44 [Challenging] • Target: Nonresponse and Response Bias • Cross-Lesson: 3.03 The Normal Distribution

A police officer in uniform asks high school students face-to-face if they have ever consumed alcohol. Which bias will most likely distort the findings?

a. Response bias, as students are likely to lie to authority figures ✓

b. Selection bias

c. Voluntary response bias

d. Cluster bias

Response bias occurs when the interviewer, phrasing, or social desirability influences respondents to provide inaccurate or untruthful answers.

45 [Challenging] • Target: Parameters vs. Statistics • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In statistical notation, which symbol represents a population parameter rather than a sample statistic?

a. μ ✓

b. $\bar{x}$

c. s

d. $\hat{p}$

μ represents the true population mean (a parameter). $\bar{x}, s, \hat{p}$ are sample statistics calculated from sample data.

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